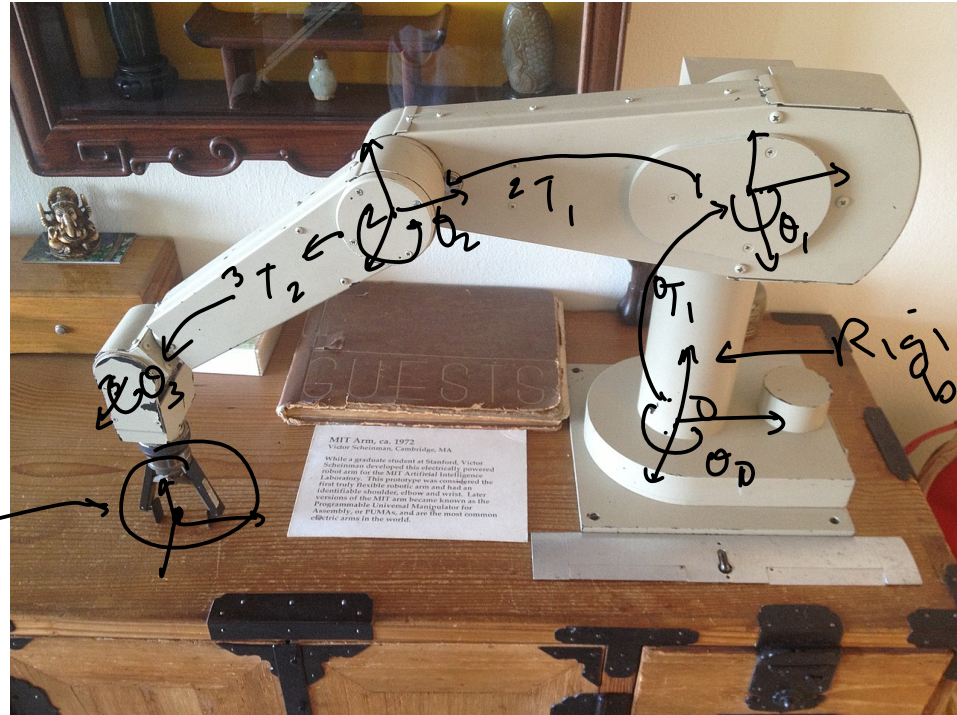


Forward and inverse kinematics



end effector

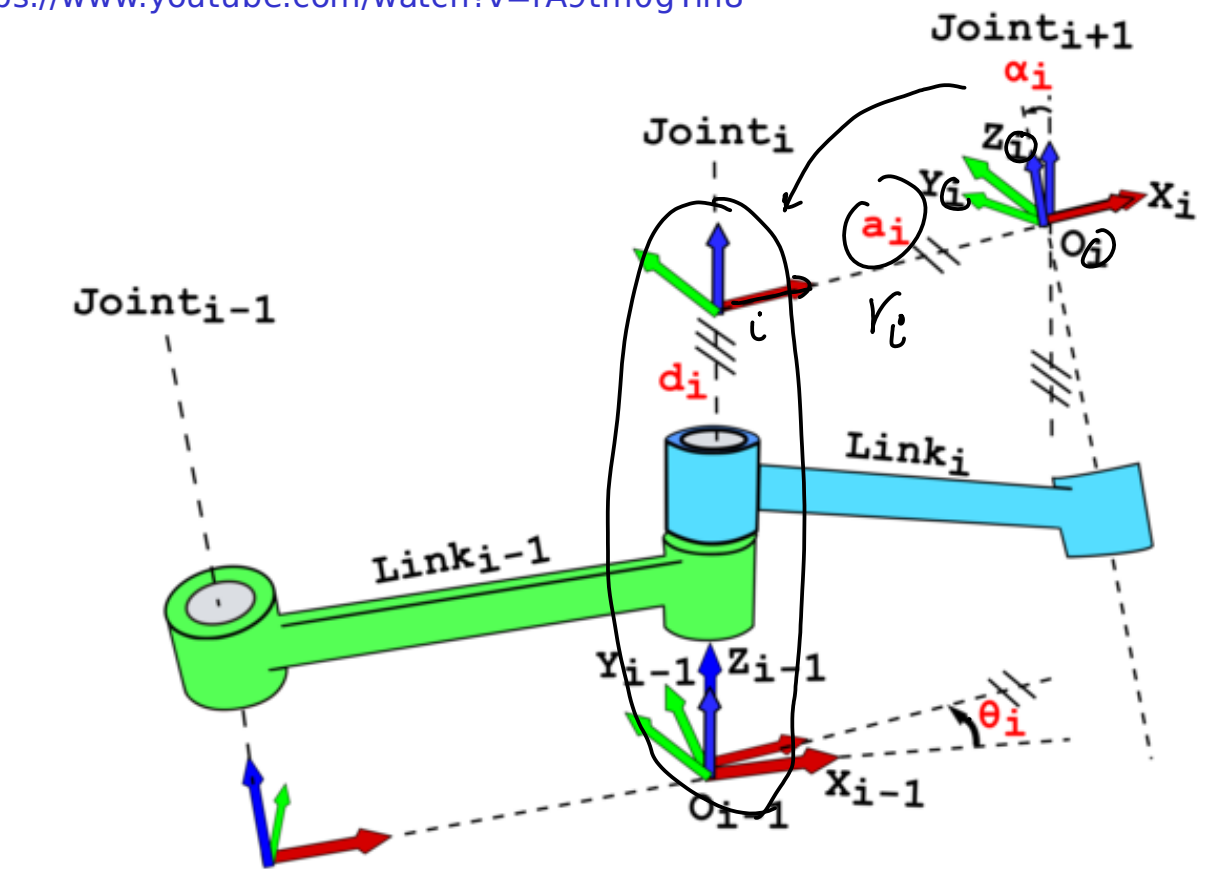
$${}^4T_0 = {}^4T_3(\theta_3) {}^3T_2(\theta_2) {}^2T_1(\theta_1) {}^1T_0(\theta_0)$$

Pose of end effector $\xleftarrow{\text{Fwd Kinematics}}$ Joint angles

Pose of end effector $\xrightarrow{\text{Inverse Kinematics}}$

Denavit Hartenberg parameters

<https://www.youtube.com/watch?v=rA9tm0gTln8>



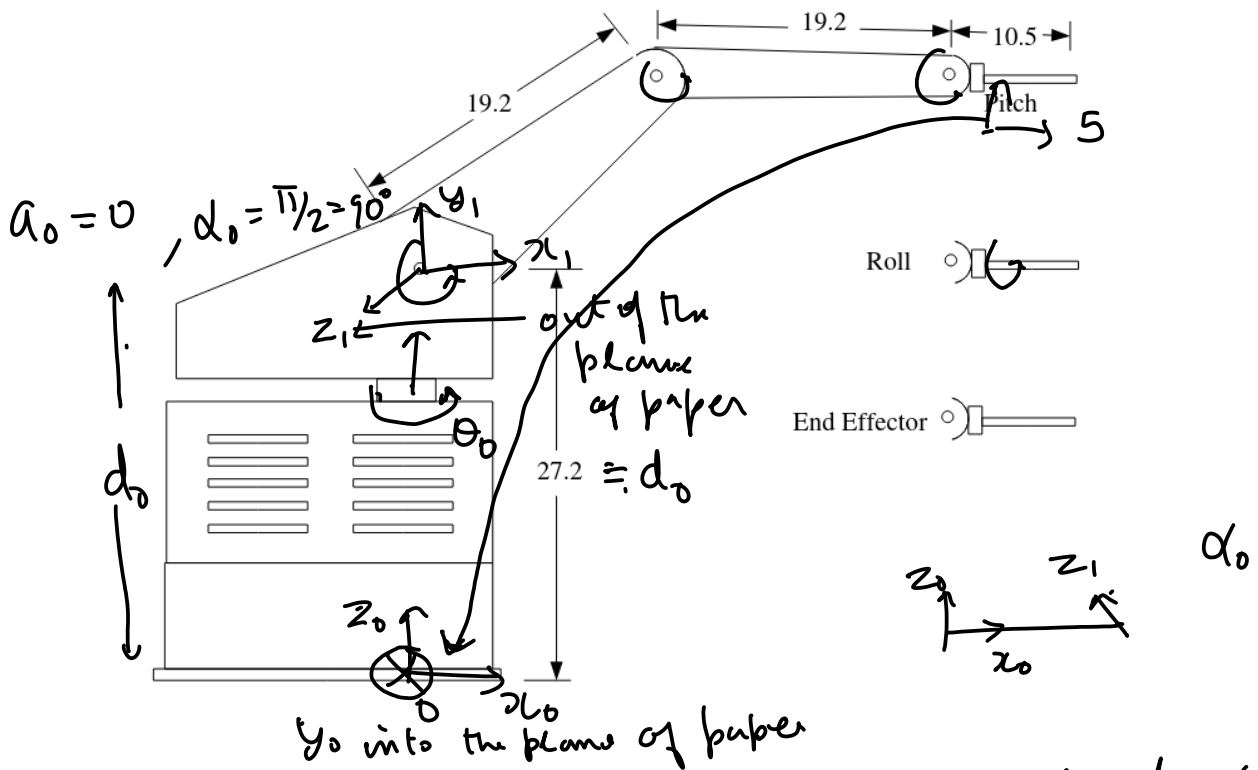
$${}^{i-1}T_i = {}^{i-1}T_{z_i} {}^{z_i}T_{x_i}$$

$$= \begin{bmatrix} R_z(\theta_i) & \underline{d_i} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_x(\alpha_i) & \underline{a_i} \\ 0 & 1 \end{bmatrix}$$

$${}^{i-1}T_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 \\ \sin \theta_i & \cos \theta_i & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & 0 \\ \sin \theta_i & \cos \theta_i & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & \cos \alpha_i & -\sin \alpha_i & 0 \\ 0 & \sin \alpha_i & \cos \alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

${}^{i-1}T_i$ ← D-H params $(\theta_i, d_i, \alpha_i, a_i)$

Find the Denavit Hartenberg Parameters of the Lab volt arm

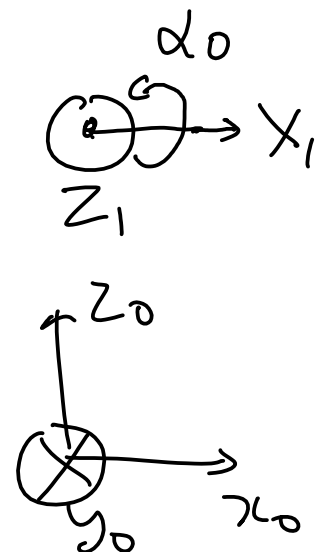


Pose = Position + orientation of the end effector (5)

$${}^0T_5 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5$$

Find DH params of the arm

i	d_i	θ_i	r_i/a_i	α_i
0	27.2	var	0	$\pi/2 = 90^\circ$
1	0	var	19.2	0
2	0	var	19.2	0
3				



α_0 is the rotation around x_0 so that

Z_0 aligns with Z_1

$$\underline{\theta}_{t+1} = \underline{\theta}_t - \alpha_t \underline{\Delta \theta}_t$$

$$\underset{\Delta \theta}{\text{minimize}} \quad \left\| \underline{r}(\underline{\theta}_{t+1}) \right\|_2^2$$

$$\underset{\Delta \theta}{\text{minimize}} \approx \left\| \underline{r}(\underline{\theta}_t) + \underbrace{\frac{\partial \underline{r}}{\partial \underline{\theta}}(\underline{\theta}_{t+1} - \underline{\theta}_t)}_{\parallel_0} + \dots \right\|_2^2$$

$$\underline{r}(\underline{\theta}_t): \mathbb{R}^n \rightarrow \mathbb{R}^m \leftarrow \begin{matrix} 12 = 9+3 \\ \uparrow \mathbb{R} \quad \uparrow \underline{\theta}_t \end{matrix}$$

vector-valued
vector function

DOF of the sum

$$\frac{\partial \underline{r}}{\partial \underline{\theta}_t} = \begin{bmatrix} \frac{\partial r_1}{\partial \theta_1} & \frac{\partial r_1}{\partial \theta_2} & \dots & \frac{\partial r_1}{\partial \theta_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial r_m}{\partial \theta_1} & \frac{\partial r_m}{\partial \theta_2} & \dots & \frac{\partial r_m}{\partial \theta_n} \end{bmatrix}_{m \times n}$$

Jacobian matrix

$$\underset{\Delta \theta_t}{\text{minimize}} \left\| \underbrace{r(\theta_t)} + \underbrace{\frac{\partial r}{\partial \theta_t}(\Delta \theta_t)} \right\|_2^2$$

$$\underset{\underline{x}}{\text{minimize}} \left\| \underline{b} - A \underline{x} \right\|_2^2$$

$$\underline{x} = \underline{A}^{\dagger} \underline{b} \quad \leftarrow \text{dagger symbol}$$

Moore-Penrose Pseudo inverse

For tall matrix $m \geq n$

$$m \begin{bmatrix} n \\ \end{bmatrix}$$

$$\underline{x} = \underbrace{(A^T A)^{-1} A^T}_{A^{\dagger}} \underline{b}$$

$$\Delta \theta_t = \left[\frac{\partial r}{\partial \theta} \right]^{\dagger} r(\theta_t)$$

Gauss Newton

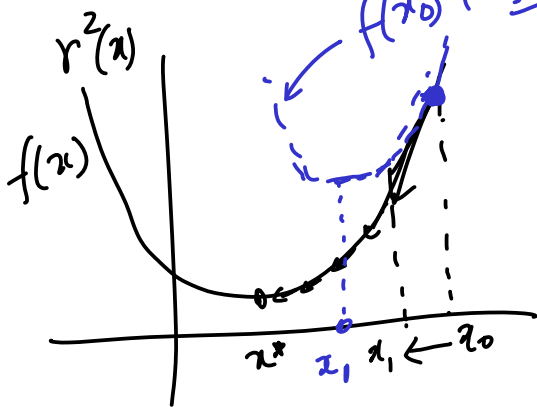
(1) Choose random starting point θ_0

(2) Repeat untill convergence

$$\theta_{t+1} = \theta_t - \left[\frac{\partial r}{\partial \theta} \right]^{\dagger} r(\theta_t)$$

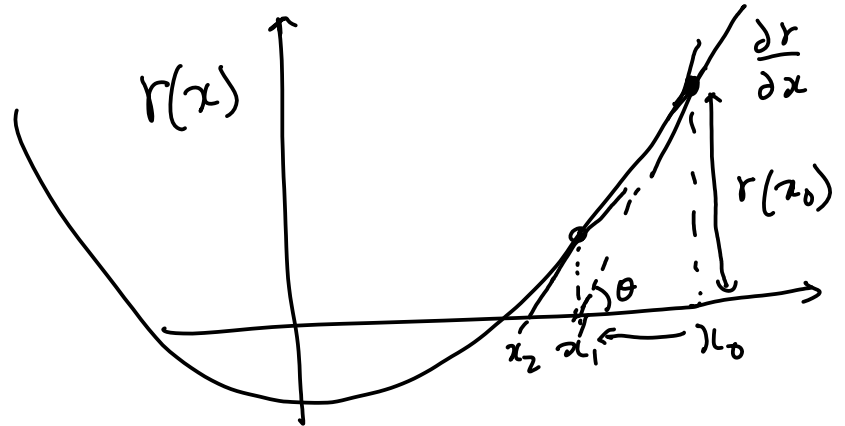
In scalar domain : Newton's method of root finding

Gradient Descent



$$x_1 = x_0 - \frac{f'(x)}{f''(x)}$$

Newton's Method



$$x_1 = x_0 - \frac{r(x_0)}{\frac{\partial r}{\partial x}}$$

$$x_1 = x_0 - \frac{r(x_0)}{r'(x_0)}$$

$$\underline{x}_1 = \underline{x}_0 - \left[\frac{\partial \underline{r}}{\partial \underline{x}}(\underline{x}) \right]^T \underline{r}(\underline{x})$$

Gauss Newton

Forward kinematics

${}^0T_4(\theta_1, \theta_2, \theta_3, \theta_4)$
end-effector pose

${}^0T_1(\theta_1) {}^1T_2(\theta_2) {}^2T_3(\theta_3) {}^3T_4(\theta_4)$
Transf. matrices
Joint angle

$$\frac{\partial {}^0T_n(\underline{\theta})}{\partial \theta_1} = \left[\frac{\partial {}^0T_1(\theta_1)}{\partial \theta_1} \right] {}^1T_2(\theta_2) {}^2T_3(\theta_3) {}^3T_4(\theta_4)$$

$$\frac{\partial}{\partial \theta_1} {}^0T_1(\theta_1) = \frac{\partial}{\partial \theta_1} \begin{bmatrix} R_z(\theta_1) & \underline{t}_z \\ \underline{0}^T & 1 \end{bmatrix} \begin{bmatrix} R_x(\alpha) & \underline{t}_x \\ \underline{0}^T & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial}{\partial \theta_1} R_z(\theta_1) & \frac{\partial}{\partial \theta_1} \underline{t}_z \\ \frac{\partial}{\partial \theta_1} \underline{0}^T & \frac{\partial}{\partial \theta_1} 1 \end{bmatrix} \begin{bmatrix} R_x(\alpha) & \underline{t}_x \\ \underline{0}^T & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial}{\partial \theta_1} R_z(\theta_1) & \frac{\partial}{\partial \theta_1} \underline{t}_z \\ \underline{0}^T & 0 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \theta_1} R_z(\theta_1) & 0 \\ \underline{0}^T & 0 \end{bmatrix} \begin{bmatrix} R_x(\alpha) & \underline{t}_x \\ \underline{0}^T & 1 \end{bmatrix}$$

From Rodrigues rotation formula

$$R(\theta, K) = I_{3 \times 3} + \sin \theta K + (1 - \cos \theta) K^2$$

$$\text{where } K = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}, \quad \hat{k} = \begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix}, \quad \|\hat{k}\| = 1$$

$$\frac{\partial}{\partial \theta} R(\theta, K) = R(\theta, K) K$$

check if this
is correct
or Prove it

You will have to use $K^3 = -K$

$$\begin{aligned} \text{L.H.S} = \frac{\partial}{\partial \theta} R(\theta, K) &= \frac{\partial}{\partial \theta} \left[I + \sin \theta K + (1 - \cos \theta) K^2 \right] \\ &= 0_{3 \times 3} + \underbrace{\cos \theta K}_{3 \times 3} + \sin \theta K^2_{3 \times 3} \end{aligned}$$

$$\begin{aligned} \text{R.H.S} = R(\theta, K) K &= \left[I + \sin \theta K + (1 - \cos \theta) K^2 \right] K \\ &= K + \sin \theta K^2 + (1 - \cos \theta) K^3 \\ &= \cancel{K} + \sin \theta K^2 - \cancel{K} + \cos \theta K \end{aligned}$$

$$\text{L.H.S} = \text{R.H.S}$$

Hence Proved

$$\frac{\partial R(\theta, K)}{\partial \theta} = R(\theta, K) K = K R(\theta, K)$$

$$R(\theta, K) = \exp(\theta K)$$

↑
Matrix exponential

⏟ Lie algebra

$$\frac{\partial}{\partial \theta_1} {}^0T_4(\theta) = \begin{bmatrix} K_z R_z(\theta_1) & 0 \\ \boxed{0_{1 \times 3}^T} & 0 \end{bmatrix} \begin{bmatrix} R_x(\alpha) & \underline{t}_x \\ 0^T & 1 \end{bmatrix} {}^1T_2(\theta_2) {}^2T_3(\theta_3) {}^3T_4(\theta_4)$$

$$\frac{\partial}{\partial \theta_2} {}^0T_4(\underline{\theta}) = {}^0T_1(\theta_1) \begin{bmatrix} K_2 R_2(\theta_2) & 0 \\ 0_{1 \times 3}^T & 0 \end{bmatrix} \begin{bmatrix} R_2(\alpha) & \underline{t}_x \\ 0^T & 1 \end{bmatrix} {}^2T_3(\theta_3) {}^3T_4(\theta_4)$$

$$\underline{r}(\underline{\theta}) = \text{vec} \left({}^0T_4(\underline{\theta}) - {}^0T_4^* \right)$$

$$\text{vec}({}^0T_4)_z = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}_{16 \times 1}$$

$$\frac{\partial \underline{r}(\underline{\theta})}{\partial \underline{\theta}} = \begin{bmatrix} \frac{\partial \underline{r}}{\partial \theta_1} & \frac{\partial \underline{r}}{\partial \theta_2} & \dots & \frac{\partial \underline{r}}{\partial \theta_4} \\ \hline 1 & 1 & \dots & 1 \end{bmatrix}$$

$\uparrow \eta = 0$
 16×4
 $\downarrow$
 12×4